

Description Logics and the Semantic Web

Umberto Straccia

ISTI - CNR,

<http://gaia.isti.cnr.it/~straccia>

straccia@isti.cnr.it

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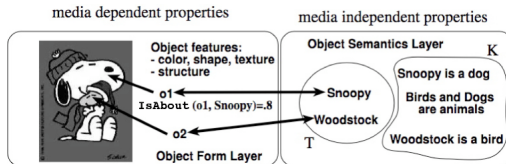
Exercise

Problem: Using Protege, try to encode the web site `www.autos.com` as ontology, starting from the lecture slides “Part III”

Solution: A partial solution is give in the file “autos.owl”

Exercise

Problem: Encode the following scenario in Description Logics and write the query “find images having an image region, which is about animals” as concept



Solution:

$$\begin{aligned} \text{Image} \sqsubseteq & (\exists \text{hasColor}.\text{Color}) \\ & \sqcap (\exists \text{hasShape}.\text{Shape}) \\ & \sqcap (\exists \text{hasTextures}.\text{Texture}) \\ & \sqcap (\exists \text{hasImageRegion}.\text{ImageRegion}) \end{aligned}$$

$$\text{Bird} \sqsubseteq \text{Animal} \quad \text{Dog} \sqsubseteq \text{Animal} \quad \text{Animal} \sqsubseteq \text{LivingThing}$$

snoopy:Dog
woodstock:Bird

$$\begin{aligned} (o_1, \text{snoopy}):&\text{IsAbout} & (o_2, \text{woodstock}):&\text{IsAbout} & o_1:&\text{ImageRegion} \\ o_2:&\text{ImageRegion} & i:&\text{Image} & (i, o_1):&\text{hasImageRegion} \\ (i, o_2):&\text{hasImageRegion} \end{aligned}$$

$$\text{QueryConcept} = \text{Image} \sqcap (\exists \text{hasImageRegion} . (\exists \text{IsAbout} . \text{Animal}))$$

Exercise

Problem: Try to encode in description logics the following definitions

- 1 Pizza, Topping and PizzaBase are disjunct concepts/classes
- 2 A cheese pizza is defined as a pizza, having a topping and having a pizza base. The topping is a cheese topping, while the base is a PizzaBase. A cheese topping is a topping.
- 3 The domain of the relation topping is pizza, while the range is topping

Solution:

- 1 We use three inclusion axioms:

$$Pizza \sqsubseteq \neg Topping$$

$$Pizza \sqsubseteq \neg PizzaBase$$

$$Topping \sqsubseteq \neg PizzaBase$$

- 2

$$CheesePizza = Pizza$$

$$\sqcap (\exists hasTopping. CheeseTopping)$$

$$\sqcap (\exists hasPizzaBase. PizzaBase)$$

$$CheeseTopping \sqsubseteq Topping$$

- 3 We use the following axioms. The first defines the domain, the second defines the range.

$$\exists hasTopping. \top \sqsubseteq Pizza$$

$$\top \sqsubseteq \forall hasTopping. PizzaTopping$$



Exercise

Problem: Translate the following Description Logic knowledge base into First-Order-Logic

<i>Pizza</i>	$\sqsubseteq$	$\neg \textit{Topping}$
<i>Pizza</i>	$\sqsubseteq$	$\neg \textit{PizzaBase}$
<i>Topping</i>	$\sqsubseteq$	$\neg \textit{PizzaBase}$
<i>CheesePizza</i>	$=$	<i>Pizza</i>
		$\sqcap (\exists \textit{hasTopping}. \textit{CheeseTopping})$
		$\sqcap (\exists \textit{hasPizzaBase}. \textit{PizzaBase})$
<i>CheeseTopping</i>	$\sqsubseteq$	<i>Topping</i>
$\exists \textit{hasTopping}. \top$	$\sqsubseteq$	<i>Pizza</i>
$\top$	$\sqsubseteq$	$\forall \textit{hasTopping}. \textit{PizzaTopping}$

Solution:

$\forall x \textit{Pizza}(x)$	$\Rightarrow$	$\neg \textit{Topping}(x)$
$\forall x \textit{Pizza}(x)$	$\Rightarrow$	$\neg \textit{PizzaBase}(x)$
$\forall x \textit{Topping}(x)$	$\Rightarrow$	$\neg \textit{PizzaBase}(x)$
$\forall x \textit{CheesePizza}(x)$	$\Leftrightarrow$	$(\textit{Pizza}(x)$
		$\wedge (\exists y \textit{hasTopping}(x, y) \wedge \textit{CheeseTopping}(y))$
		$\wedge (\exists y \textit{hasPizzaBase}(x, y) \wedge \textit{PizzaBase}(x, y)))$
$\forall x \textit{CheeseTopping}(x)$	$\Rightarrow$	<i>Topping</i> (x)
$\forall x \exists y \textit{hasTopping}(x, y) \wedge \textit{True}(y)$	$\Rightarrow$	<i>Pizza</i> (x)
$\forall x \textit{True}(x)$	$\Rightarrow$	$(\forall y \textit{hasTopping}(x, y) \Rightarrow \textit{PizzaTopping}(y))$

Exercise

Problem: Translate the following Description Logic knowledge base into LISP-like syntax of the reasoner fuzzyDL
<http://gaia.isti.cnr.it/~straccia/software/fuzzyDL/fuzzyDL.html>

<i>Pizza</i>	$\sqsubseteq$	$\neg \textit{Topping}$
<i>Pizza</i>	$\sqsubseteq$	$\neg \textit{PizzaBase}$
<i>Topping</i>	$\sqsubseteq$	$\neg \textit{PizzaBase}$
<i>CheesePizza</i>	$=$	$\textit{Pizza} \sqcap (\exists \textit{hasTopping}.\textit{CheeseTopping})$ $\sqcap (\exists \textit{hasPizzaBase}.\textit{PizzaBase})$
<i>CheeseTopping</i>	$\sqsubseteq$	<i>Topping</i>
$\exists \textit{hasTopping}.\top$	$\sqsubseteq$	<i>Pizza</i>
$\top$	$\sqsubseteq$	$\forall \textit{hasTopping}.\textit{PizzaTopping}$

p:Pizza t:CheeseTopping pb:PizzaBase (p, t):hasTopping (p, pb):hasPizzaBase

Solution:

(disjoint Pizza Topping PizzaBase)

```
(define-concept CheesePizza (and Pizza
  (some hasTopping CheeseTopping)
  (some hasPizzaBase PizzaBase)))
```

(implies CheeseTopping Topping)

```
(domain hasTopping Pizza)
(range hasTopping PizzaTopping)
```

```
(instance p Pizza) (instance t CheeseTopping) (instance pb PizzaBase)
(related p t hasTopping) (related p pb hasPizzaBase)
```

Exercise

Problem: Verify that the individual “p” of the previous example is a cheese pizza

Solution: Submit the following file to the fuzzyDL reasoner, by selecting the reasoning mode “classical”
(disjoint Pizza Topping PizzaBase)

```
(define-concept CheesePizza (and Pizza
                                   (some hasTopping CheeseTopping)
                                   (some hasPizzaBase PizzaBase)))

(implies CheeseTopping Topping)

(domain hasTopping Pizza)
(range hasTopping PizzaTopping)

(instance p Pizza) (instance t CheeseTopping) (instance pb PizzaBase)
(related p t hasTopping) (related p pb hasPizzaBase)

(individual-instance? p CheesePizza)
```

